A FINITE SET OF POINTS WITH MATCHING LOCAL RINGS ON AN AFFINE SOURCE LIES IN AN AFFINE OPEN OF THE QUASI-SEPARATED TARGET

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites six cross-chapter labels, each recorded in assembly.external_references with its canonical tag as a navigation pointer.

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Allowed standard prerequisites (editorial).

- Criterion for affineness via a limit presentation. Stacks Project, Limits of Schemes, Proposition limits-proposition-affine (tag 01Z2). Provides the affine-open criterion on which the whole proof rests.
- Local rings and generic points under a morphism. Stacks Project, Schemes (generic points and local rings of schemes). Lets the hypotheses on T^0 -> X^0 and the local rings be used to control the points x_j.

Source reference. Original reference 05YU: sources/stacks-chow/tags.txt, source lines 7668–7668. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 09MV: sources/stacks-chow/tags.txt, source lines 12193–12193. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01Z6: sources/stacks-chow/tags.txt, source lines 2581–2581. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 09NH: sources/stacks-chow/tags.txt, source lines 12215–12215. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 09NG: sources/stacks-chow/tags.txt, source lines 12214–12214. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01ZY: sources/stacks-chow/tags.txt, source lines 2613–2613. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $f : T \to X$ be a morphism of schemes. Let X^0, resp. T^0 denote the sets of generic points of irreducible components. Let $t_1, \ldots, t_m \in T$ be a finite set of points with images $x_j = f(t_j)$. If*

(1) *T is affine,*
(2) *X is quasi-separated,*

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(3) X^0 *is finite*
(4) $f(T^0) \subset X^0$ *and* $f : T^0 \to X^0$ *is injective, and*
(5) $\mathcal{O}_{X,x_j} = \mathcal{O}_{T,t_j}$,

then there exists an affine open of X *containing* $x_1, \ldots, x_r$.

Proof. Using Limits, Proposition 05YU there is an immediate reduction to the case where X and T are reduced. Details omitted.

Assume X and T are reduced. We may write $T = \lim_{i \in I} T_i$ as a directed limit of schemes of finite presentation over X with affine transition morphisms, see Limits, Lemma 09MV. Pick $i \in I$ such that T_i is affine, see Limits, Lemma 01Z6. Say $T_i = \mathrm{Spec}(R_i)$ and $T = \mathrm{Spec}(R)$. Let $R' \subset R$ be the image of $R_i \to R$. Then $T' = \mathrm{Spec}(R')$ is affine, reduced, of finite type over X, and $T \to T'$ dominant. For $j = 1, \ldots, r$ let $t'_j \in T'$ be the image of t_j. Consider the local ring maps

$$\mathcal{O}_{X,x_j} \to \mathcal{O}_{T',t'_j} \to \mathcal{O}_{T,t_j}$$

Denote $(T')^0$ the set of generic points of irreducible components of T'. Let $\xi \rightsquigarrow t'_j$ be a specialization with $\xi \in (T')^0$. As $T \to T'$ is dominant we can choose $\eta \in T^0$ mapping to ξ (warning: a priori we do not know that η specializes to t_j). Assumption (3) applied to η tells us that the image θ of ξ in X corresponds to a minimal prime of $\mathcal{O}_{X,x_j}$. Lifting ξ via the isomorphism of (5) we obtain a specialization $\eta' \rightsquigarrow t_j$ with $\eta' \in T^0$ mapping to $\theta \rightsquigarrow x_j$. The injectivity of (4) shows that $\eta = \eta'$. Thus every minimal prime of $\mathcal{O}_{T',t'_j}$ lies below a minimal prime of $\mathcal{O}_{T,t_j}$. We conclude that $\mathcal{O}_{T',t'_j} \to \mathcal{O}_{T,t_j}$ is injective, hence both maps above are isomorphisms.

By Lemma 09NH there exists an open $U \subset T'$ containing all the points t'_j such that $U \to X$ is a local isomorphism as in Lemma 09NG. By that lemma we see that $U \to X$ is an open immersion. Finally, by Properties, Lemma 01ZY we can find an open $W \subset U \subset T'$ containing all the t'_j. The image of W in X is the desired affine open. $\square$